

GF(27) in GALG and FFGFT:

Why $G(3)$ Contains No GF(27) but $G(6)$ Does

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1 September 2026

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GF(27) in GALG and FFGFT

Abstract

Doug Matzke reports in his IPI mail of 1 September 2026 that a systematic search in GALG $G(3)$ (6561 attempts) finds no element of order 26. This document explains algebraically why, shows where such elements exist in $G(6)$, and draws the consequences for the FFGFT mass layer.

Three theorems: **(A)** $G(3) \cong M_2(\text{GF}(9))^2$: order 26 is algebraically impossible in $G(3)$ — not a search error, but a theorem. **(B)** $G(6) \cong M_8(\text{GF}(9))$: order-26 elements exist; a 5-blade example can be verified directly in GALG. **(C)** $\text{GF}(27)$ is present in $G(6)$ as an order structure, not as a subfield (since $3 \nmid 8$) — exactly the FFGFT statement from Doc. 338.

Consequence: $1/\alpha = 3700/27$ lives on the $G(3)/\text{GF}(9)$ level; the lepton mass layer $(m_\mu/m_e)^2 = 43200$ needs $G(6)$ with order-26 elements. Verification script: `pruef_341_gf27_in_galg.py` (all assertions **[B]**).

.1 Background and Notation

This section provides the necessary background for readers not immediately familiar with all the algebraic structures involved. It can be skipped by specialists.

Status markers

This document uses the standard FFGFT corpus status markers:

- **[K]** — *Confirmed*: algebraically or numerically fully proved; all assertions in the verification script passed.
- **[B]** — *Established*: numerically secured by sampling or explicit construction; analytic proof present or straightforward.
- **[S]** — *Speculative*: indication or conjecture, not yet fully proved.

Galois fields $\text{GF}(p^n)$

A *Galois field* $\text{GF}(p^n)$ is the unique finite field with p^n elements, where p is prime (the *characteristic*) and $n \geq 1$. Its multiplicative group $\text{GF}(p^n)^* = \text{GF}(p^n) \setminus \{0\}$ is cyclic of order $p^n - 1$.

- $\text{GF}(3)$: the integers $\{0, 1, 2\}$ modulo 3. Multiplicative group order 2.
- $\text{GF}(9) = \text{GF}(3^2)$: elements $a + bi$ with $a, b \in \{0, 1, 2\}$ and $i^2 = -1 \equiv 2$. Multiplicative group order 8. This is Doug's symmetric Z3C.
- $\text{GF}(27) = \text{GF}(3^3)$: 27 elements, multiplicative group order $26 = 2 \cdot 13$.
- $\text{GF}(81) = \text{GF}(3^4)$: 81 elements, multiplicative group order $80 = 2^4 \cdot 5$.

The *order* of an element x in a group is the smallest positive k with $x^k = 1$. If the group has order m , then every element order divides m (Lagrange's theorem). This is the key fact used throughout.

Clifford algebra $\text{Cl}(n)$ and $\text{GALG } \mathbf{G}(n)$

The Clifford algebra $\text{Cl}(n)$ is generated by n basis vectors $e_1, \dots, e_n$ with the rule $e_i^2 = +1$ (hyperbits in Matzke's language) and $e_i e_j = -e_j e_i$ for $i \neq j$. A *blade* of grade r is a product $e_{i_1} e_{i_2} \dots e_{i_r}$ with $i_1 < i_2 < \dots < i_r$; together with the scalar 1 (grade 0) they form a basis of 2^n elements. $\text{GALG } \mathbf{G}(n)$ is $\text{Cl}(n)$ over the coefficient field $\text{GF}(9) = \text{Z3C}$ (i.e. each blade carries a $\text{Z3C} = \text{GF}(9)$ coefficient). As a matrix algebra: $\text{Cl}(3) \cong M_2(\text{GF}(9))^2$ and $\text{Cl}(6) \cong M_8(\text{GF}(9))$, where $M_k(F)$ denotes $k \times k$ matrices over the field F .

Why $\text{GF}(27)$ and order 26?

The lepton mass ratio $(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 5^2 \cdot 27$ involves $27 = |\text{GF}(27)|$ (Doc. 338, pruef_330/331). The multiplicative group $\text{GF}(27)^*$ has order $26 = 2 \cdot 13$. An element of order 26 in any algebra generates a copy of $\text{GF}(27)^*$. Doug's search for order-26 elements in $\mathbf{G}(3)$ is therefore a direct test of whether $\text{GF}(27)$ structure is present in $\mathbf{G}(3)$.

Companion matrix

Given a monic polynomial $f(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$, its *companion matrix* is the $n \times n$ matrix with 1's on the subdiagonal and the coefficients $-a_0, \dots, -a_{n-1}$ in the last column. Its

characteristic polynomial is $f(x)$, so its eigenvalues are the roots of f and its order in GL_n equals the multiplicative order of the roots.

.2 Occasion and Starting Point

Doug Matzke (IPI, 1 Sept. 2026) reports finding in GALG $G(3)$: $GF(9)$ in the generator $(a+ja)^8 = 1$; $GF(81)$ in $G(3)$ with $(1+a+b+c+(a \wedge b))^{80} = 1$; and element orders $\{2, 3, 4, 8, 16, 32, 40, 80\}$ — all of which divide $80 = |GF(81)^*|$, consistent with the $Cl(3) \cong M_2(GF(9))^2$ structure where eigenvalues live in $GF(81)$.

An explicit search for elements of order 26 (which would signal $GF(27)^*$ structure):

`gasolve([a,b,c], is_root26): Attempted 6561 with 0 found.`

$6561 = 3^8$ is the number of elements in a 4-trit search space — exhaustive at that depth.

This is not a search error or a limitation of the tool. It is an algebraic theorem: order 26 requires the prime factor 13, and 13 does not appear in $80 = |GF(81)^*|$. The following sections prove this precisely and show where $GF(27)$ structure does appear — in $G(6)$, not $G(3)$.

.3 Theorem A: $G(3)$ Contains No $GF(27)$

Structure of $G(3)$

GALG $G(n) = Cl(n)$ over the symmetric $Z3C = GF(9) = GF(3)[i]$, where $i^2 = -1$, characteristic 3. For odd $n = 3$:

$$G(3) = Cl(3) \cong M_2(GF(9)) \oplus M_2(GF(9)) \quad (1)$$

Every irreducible representation is 2-dimensional over $GF(9)$.

Eigenvalue Structure

The eigenvalues of a 2×2 matrix over $GF(9)$ lie in the algebraic closure — in the smallest extension field $GF(9^2) = GF(81)$.

$$|GF(81)^*| = 80 = 2^4 \cdot 5 \quad (2)$$

The number 13 does not divide 80. Therefore:

- No element of $M_2(GF(9))$ has an order divisible by 13.
- Order 26 requires $13 \mid \text{ord}(X)$ — impossible in $G(3)$.
- Doug's search space $\{a, b, c\}$ spans $G(3)$ — the result 0/6561 is algebraically forced **[B]**.

Consistency with Doug's Findings

Table 1: Orders in GALG $G(3)$ and their Galois origin

Order	Origin	Divides
2	$ \text{GF}(3)^* $	80
4	$\varphi(5) = 4$	80
5	min. prime factor $ \text{GF}(81)^* $	80
8	$ \text{GF}(9)^* $	80
16, 20, 40, 80	subgroups of $\text{GF}(81)^*$	80
3	unipotent (char. 3)	—
26	$ \text{GF}(27)^* $, needs factor 13	not 80

A numerical sample of 3000 random $\text{Cl}(3)$ elements confirms: no order is divisible by 13 **[B]**.

.4 Theorem B: $G(6)$ Contains Order-26 Elements

Structure of $G(6)$

$$G(6) = \text{Cl}(6) \cong M_8(\text{GF}(9)) \quad (3)$$

Irreducible representations are 8-dimensional over $\text{GF}(9)$.

Construction via Irreducible Polynomial

The polynomial $f(x) = x^3 + 2x + 1$ over $\text{GF}(3)$ (equivalent to $x^3 - x + 1$) has no root in $\text{GF}(3)$ and is therefore irreducible. Since $\gcd(3, 2) = 1$, f remains irreducible over $\text{GF}(9)$; its roots lie in $\text{GF}(27) \setminus \text{GF}(3)$. The companion matrix

$$C_f = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \in M_3(\text{GF}(3)) \quad (4)$$

has order 26 in $\text{GL}_3(\text{GF}(3))$ (f is primitive) **[B]**.

Embedding in $G(6)$ as a GALG Element

The 8×8 element $X = C_f \oplus C_f \oplus (-I_2)$ satisfies:

- $X^{26} = 1$, $X^{13} \neq 1$, $X^2 \neq 1$ — order exactly 26 **[B]**
- Blade decomposition over 38 Z3C blades from $\{e_1, \dots, e_6\}$

Minimal Example: 5 Blades, Directly Verifiable in GALG

A systematic search shows: order-26 elements need at least 5 blades; with 4 blades there are none. The following 5-blade element can be verified directly in Doug's GALG tool:

$$X = -(1+i)(e_1e_2) + (1+i)(e_1e_2e_3e_6) + (1-i)(e_1e_2e_5e_6) + (1+i)(e_1e_4e_5e_6) - (e_1e_2e_3e_4e_6) \quad (5)$$

GALG syntax:

```

X = -(1+1j)*(e1^e2) + (1+1j)*(e1^e2^e3^e6)
+ (1-1j)*(e1^e2^e5^e6) + (1+1j)*(e1^e4^e5^e6)
- (e1^e2^e3^e4^e6)
X**26 == 1 # True
X**13 == 1 # False

```

[B] (pruef_341, all assertions passed)

Frequency in $G(6)$

Table 2: Fraction of elements with $13 \mid \text{ord}(X)$ in $G(6)$ by blade support

Support (number of blades)	Hits / Invertible	Fraction
≤ 4	0 / ~450	0 %
5	rare	< 1 %
6	85 / 478	17.8 %
10	190 / 511	37.2 %

Order-26 elements are common in $G(6)$ — Doug's search in $G(3)$ ($\text{span}\{a, b, c\}$) could not find them for two reasons: algebraically (Theorem A) and because $G(3) \subsetneq G(6)$.

.5 Theorem C: $\text{GF}(27)$ as Order Structure, not Subfield

Subfield Criterion

A unital embedding $\text{GF}(27) \hookrightarrow M_n(\text{GF}(9))$ makes $\text{GF}(9)^n$ a $\text{GF}(27) \otimes \text{GF}(9) = \text{GF}(729)$ -vector space. Since $[\text{GF}(729) : \text{GF}(9)] = 3$, we need $3 \mid n$.

Table 3: $\text{GF}(27)$ as subfield in $M_n(\text{GF}(9))$

n	$3 \mid n$	Subfield possible?
2	no	no ($G(3)$)
8	no	no ($G(6)$)
3	yes	yes
6	yes	yes
9	yes	yes

For $G(6)$ with $n = 8$: no $\text{GF}(27)$ subfield. But the minimal polynomial of X factors:

$$\chi_X(x) = (x^3 + 2x + 1)^2 \cdot (x + 1)^2 \in \text{GF}(3)[x] \quad (6)$$

Two cubic factors — eigenvalues in $\text{GF}(27)$ — and one linear factor from $\text{GF}(3)$. $\text{GF}(27)$ appears as splitting field, not as subalgebra.

Connection to Docs. 330/338

This is exactly the FFGFT statement from Docs. 330 and 338:

$$\text{GF}(9) \not\subset \text{GF}(27) \quad (\text{since } \gcd(2, 3) = 1) \quad (7)$$

Both fields sit in parallel in $\text{GF}(3^6) = \text{GF}(729)$, not in a tower. The mass-ratio identity $(m_\mu/m_e)^2 = |\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)|$ uses both fields *in parallel* — consistent with Theorem C.

.6 Consequences for FFGFT and GALG

Table 4: GF(27) structure in GALG and FFGFT

Level	GALG	FFGFT	Status
α	G(3)/GF(9)	$1/\alpha = 3700/27$, cancels	[B]
Lepton masses	G(6), ord. 26 needed	$(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 5^2 \cdot 27$	[K]
GF(27) subfield	absent in G(3), G(6)	parallel in GF(729)	[B]

The fine-structure constant $1/\alpha = 3700/27$ lives on the GF(9)/G(3) level: $|\text{GF}(27)| = 27$ cancels in the derivation. Doug's statement "I'm not working on mass" and his search result 0/6561 are both consistent with this layering:

- G(3): GF(9), GF(81), α -structure — Doug's working level **[B]**
- G(6): order-26 elements, lepton mass layer — FFGFT level **[K]**

Appendix: For the Reader Without an Algebra Background

This appendix explains the key ideas in plain language, assuming no prior knowledge.

What is a finite field?

A field is a set of numbers in which you can add, subtract, multiply and divide (except by zero). Familiar examples are the real numbers $\mathbb{R}$ and the rational numbers $\mathbb{Q}$. A *finite* field has only finitely many elements. You compute "modulo" a prime p : discard all multiples of p . Example GF(3): the numbers $\{0, 1, 2\}$ with $1 + 2 = 0$, $2 + 2 = 1$, etc.

What does $\text{GF}(p^n)$ and $\text{GF}(p^n)^*$ mean?

$\text{GF}(p^n)$ is the unique finite field with exactly p^n elements. The *star* $\text{GF}(p^n)^*$ denotes all elements *except zero* — analogous to $\mathbb{Z}^* = \{\dots, -1, +1, \dots\}$ vs. $\mathbb{Z}$:

$$|\text{GF}(27)| = 27 \quad |\text{GF}(27)^*| = 26 = 27 - 1$$

Group order and Lagrange's theorem

$\text{GF}(p^n)^*$ is always a cyclic group of order $p^n - 1$. The *order* of an element x is the smallest k with $x^k = 1$. By Lagrange's theorem, every element order *divides* $p^n - 1$. Think of a clock hand that returns every 4 hours: it cannot fit on a 6-hour clock because 4 does not divide 6.

Concretely:

- $\text{GF}(81)^*$: order $80 = 2^4 \cdot 5$. Prime factors: 2 and 5 only. Order $26 = 2 \cdot 13$ is impossible — 13 does not divide 80.
- $\text{GF}(27)^*$: order $26 = 2 \cdot 13$. Elements of order 13 and 26 exist here.

Why does GF(27) appear in the lepton masses?

$(m_\mu/m_e)^2 = 43200 = 8^2 \cdot 25 \cdot 27$. The 27 is $|\text{GF}(27)|$ (total field size including zero); the 8 is $|\text{GF}(9)^*|$ (group order of GF(9)). These arise because the FFGFT winding numbers on the T^4 torus are algebraically linked to these fields.

Why does Doug find no $\text{GF}(27)$ in $\text{G}(3)$?

$\text{G}(3)$ uses 2×2 matrices over $\text{GF}(9)$; eigenvalues live in $\text{GF}(81)$. $\text{GF}(81)^*$ has order 80, and $13 \nmid 80$, so order 26 is impossible. In $\text{G}(6)$ with 8×8 matrices, order 26 is achievable. A 5-blade example is given in §.4 and directly verifiable in GALG.

Verification Script

`pruef_341_gf27_in_galg.py` — all three theorems verified numerically ($\text{Cl}(6)$ over $\text{GF}(9)$ as 8×8 matrices; exact order determination; blade decomposition with Z3C coefficients; samples of 3000 and 600 elements). All assertions passed **[B]**.

Note on AI Assistance

This document was prepared with support from Claude (Anthropic). All algebraic computations were verified by Python scripts with exact arithmetic over $\text{GF}(9)$ as 8×8 matrix representations. The physical interpretations and conclusions are by Johann Pascher (ORCID 0009-0000-6518-4064).

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